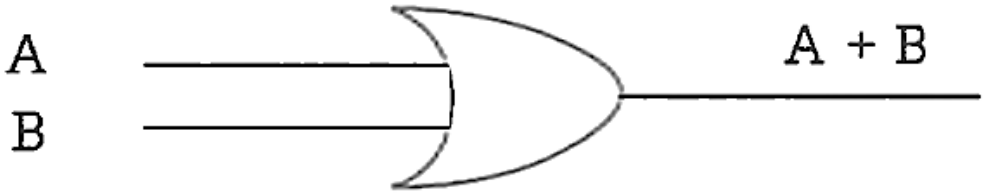
Operación de micros e interfases

Conalep San José Iturbide

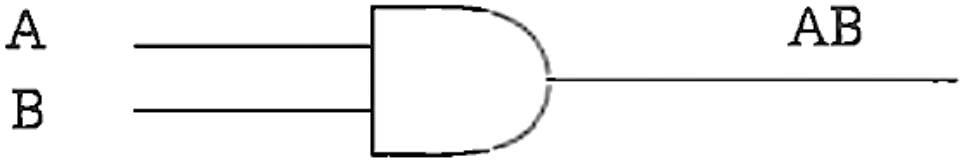
Compuertas y circuitos lógicos

Compuertas Básicas

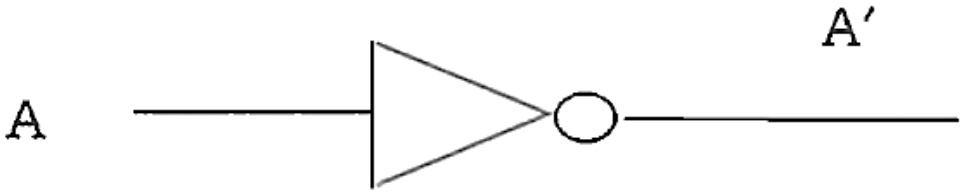
O (OR)



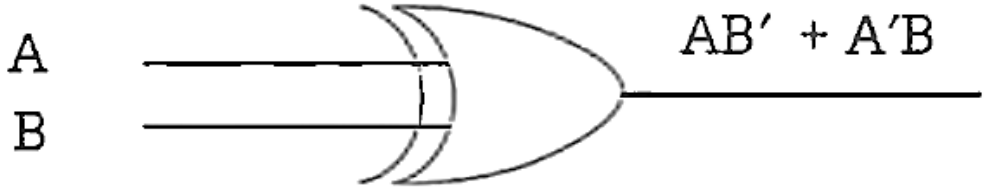
Y (AND)



NO (NOT)

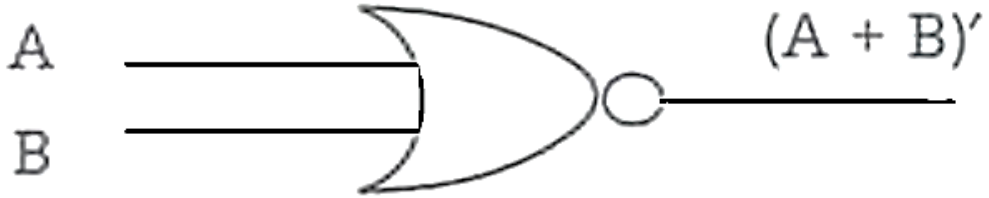


Or exclusivo (XOR)

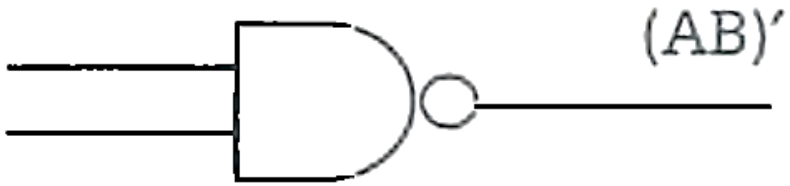


Compuertas Compuestas

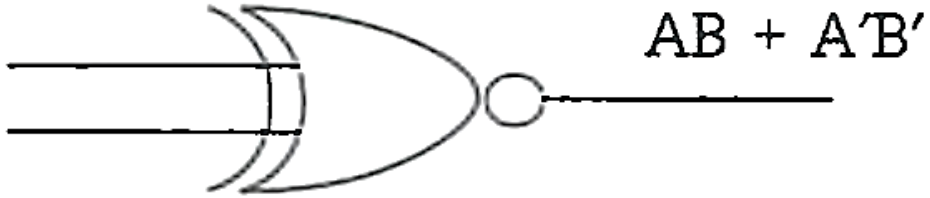
NOR



NAND



XNOR



Compuertas y circuitos lógicos

- Representar las siguientes expresiones booleanas usando compuertas lógicas básicas:

a) $F = AB' + A'C' + B$

b) $F = (A + B)' + (B + C')A$

c) $F = (AB)' + (A' + C)'$

http://weblidi.info.unlp.edu.ar/catedras/organiza/circuitos/editor_simple.html

Compuertas y circuitos lógicos

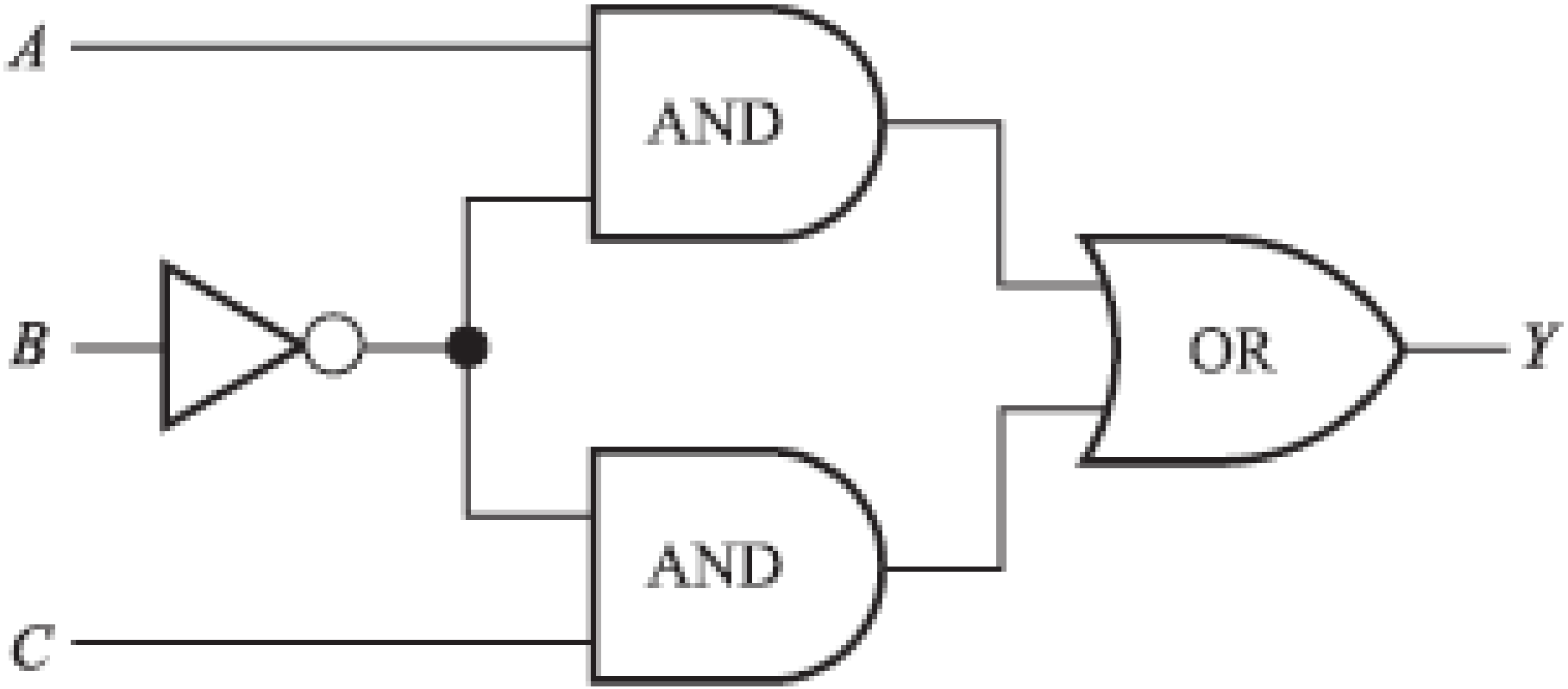
¿Cuál es el circuito de la expresión booleana F = A(B + C') + A'D hecho solo con compuertas Nand?

Simplificar las siguientes expresiones booleanas usando los teoremas del algebra booleana.

- a) F = A'B'D' + A'BD' + A'BD + ABD
- b) F = A'CD + ACD + A'B'D + A'B'C + AB'D + AB'CD'
- **c)** F = A'B'C'D' + A'B'CD' + A'BC'D + A'BCD + ABC'D' + ABCD' + AB'C'D' + AB'CD'

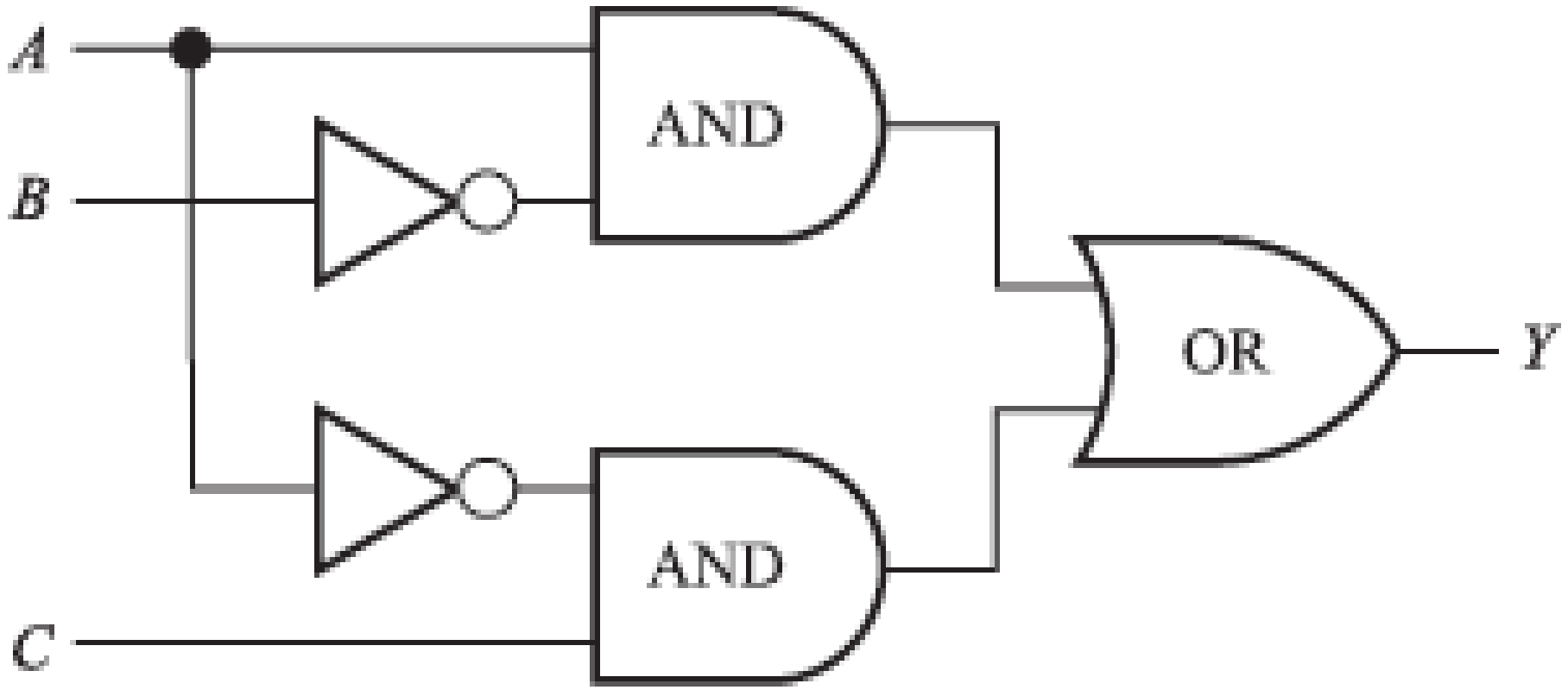
Proteus 1

Construye el siguiente circuito lógico con sus compuertas, y construye su tabla de verdad

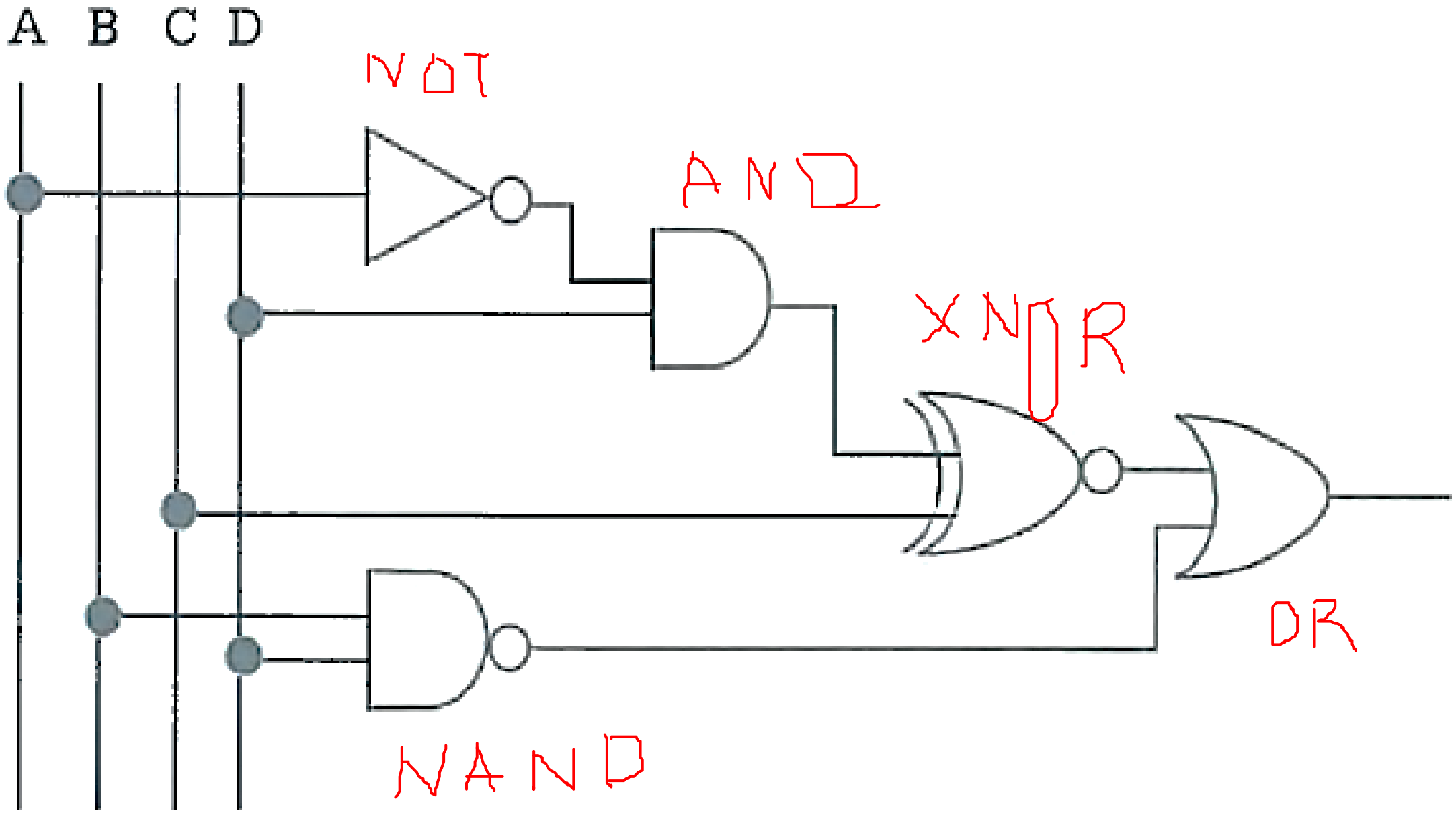


Proteus 2

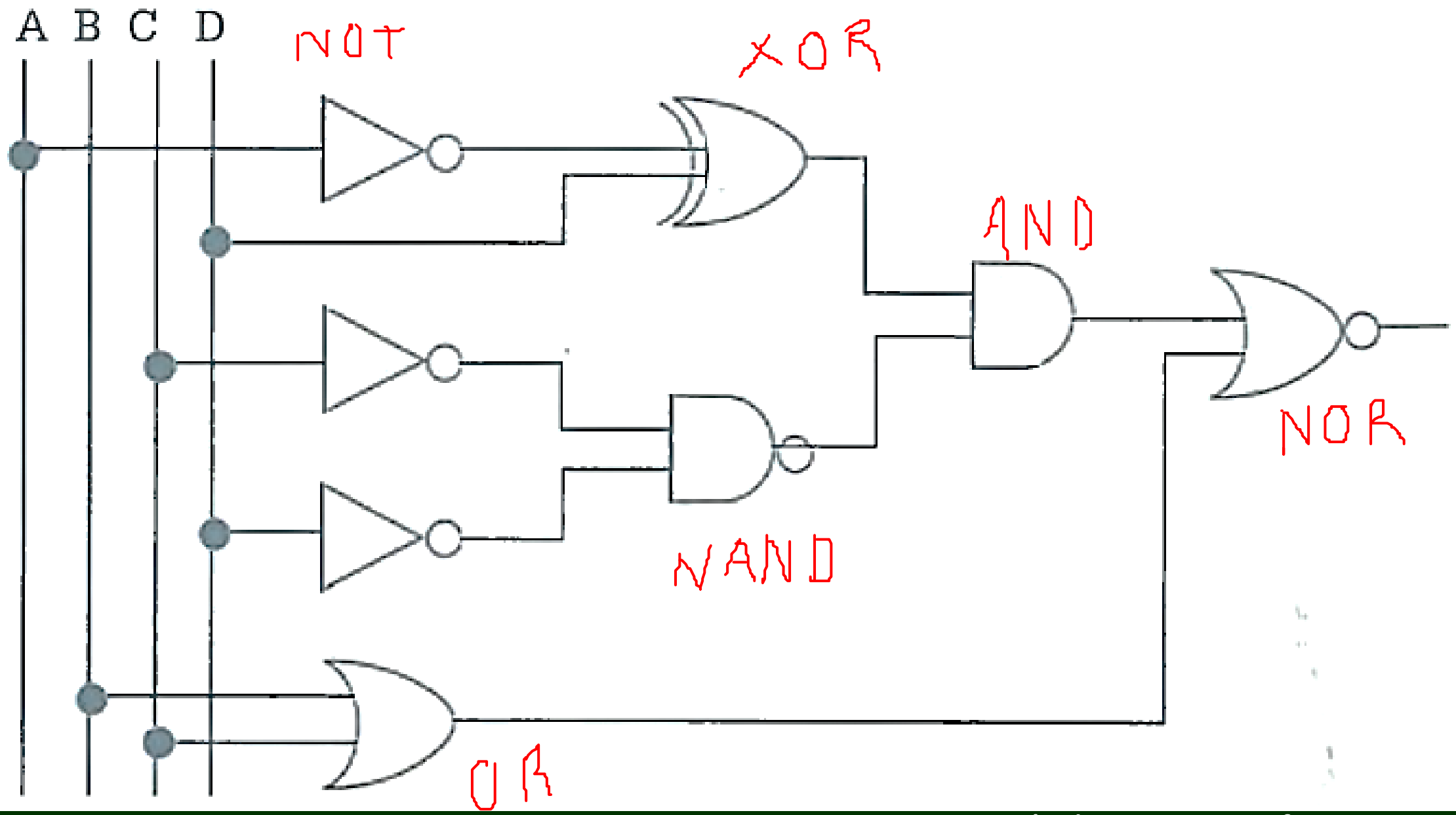
Construye el siguiente circuito lógico con sus compuertas, y construye su tabla de verdad



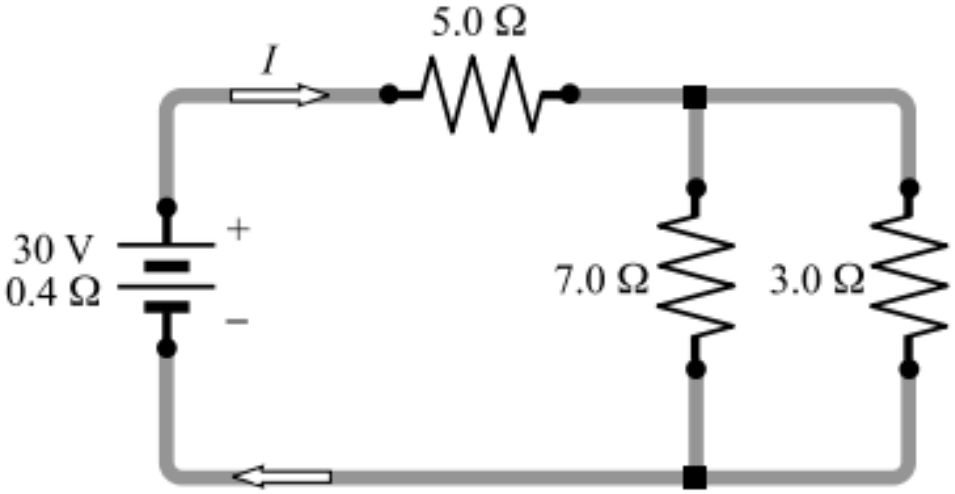
Proteus 3



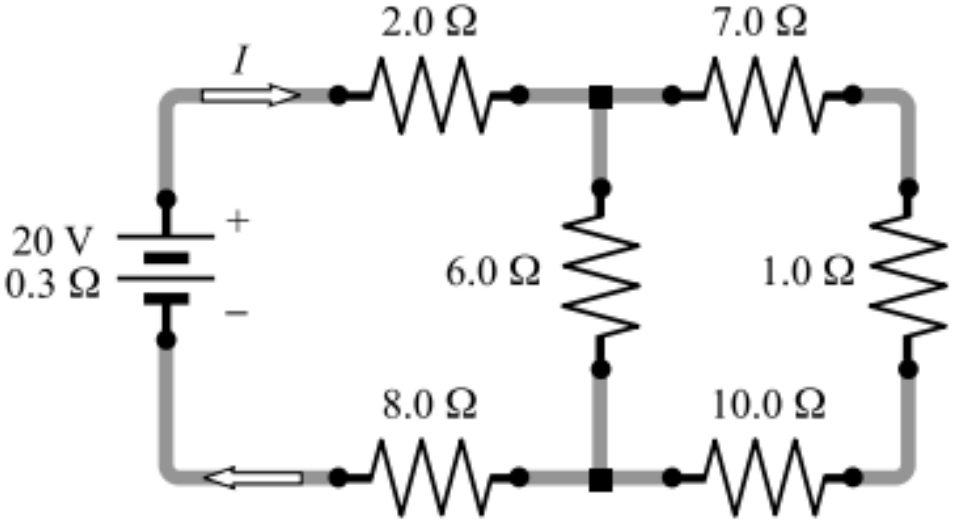
Proteus 4



Proteus 6 y 7

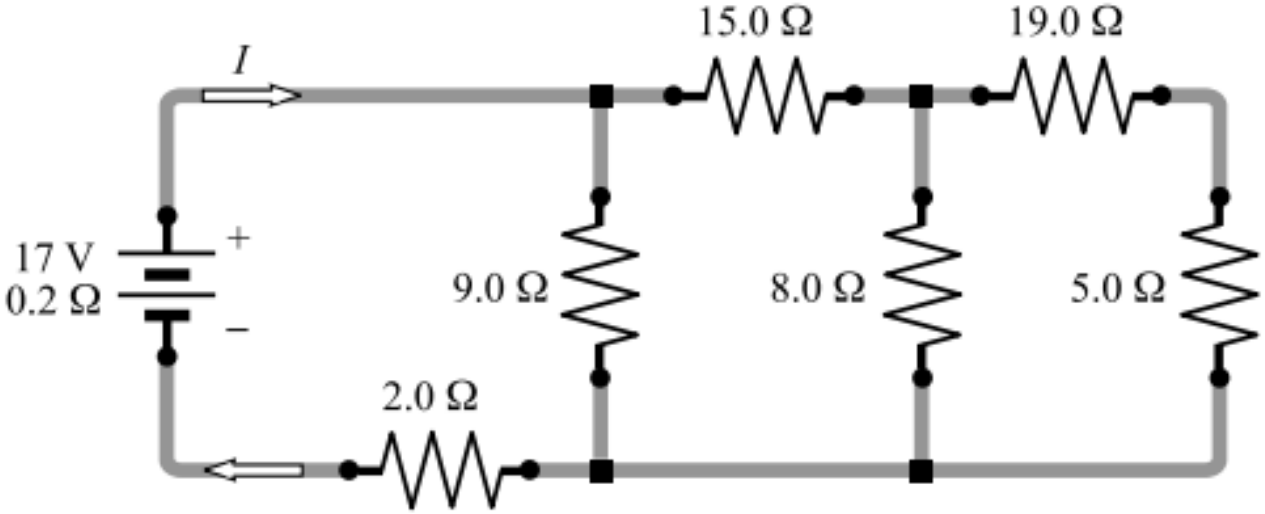


(a)



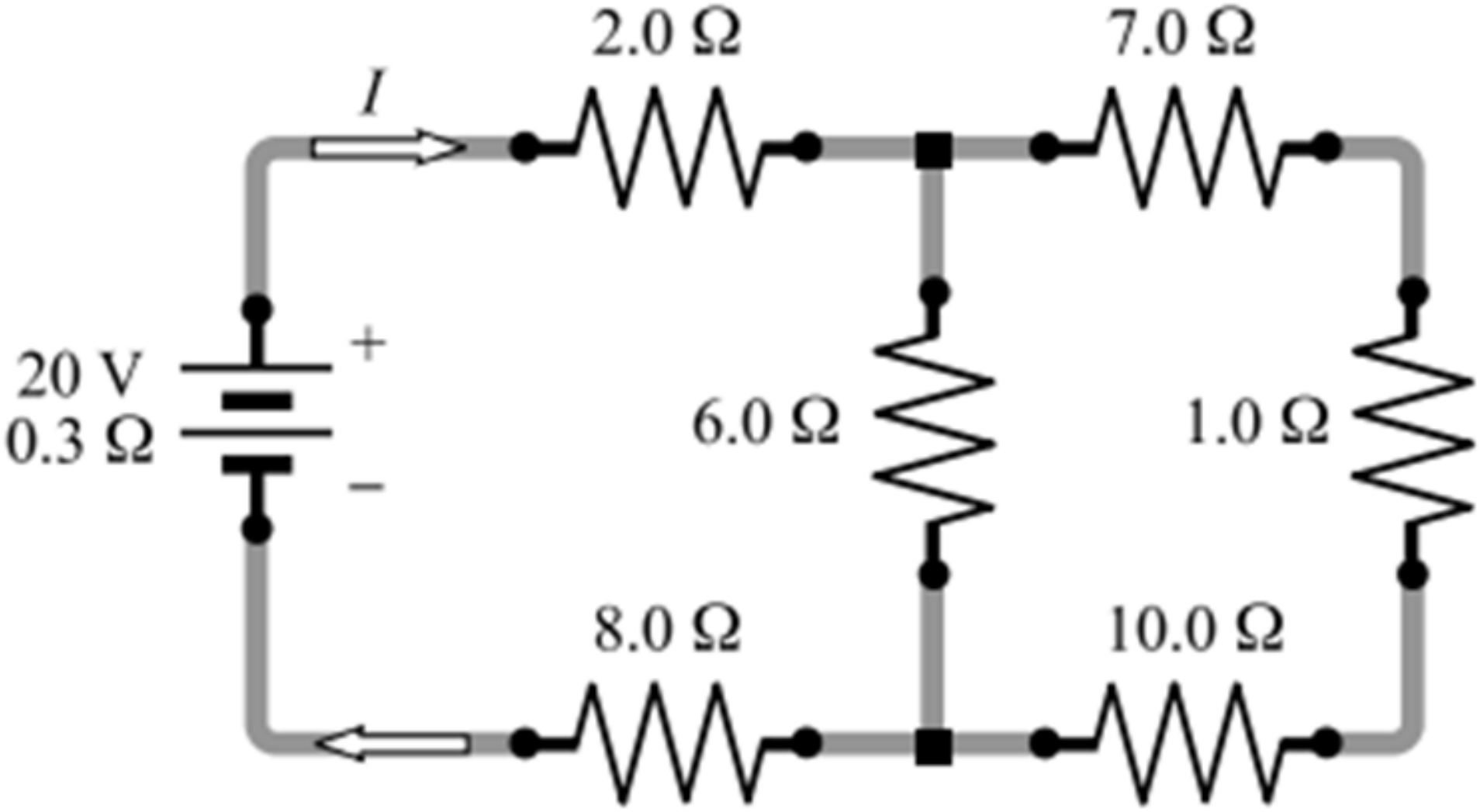
(b)

Para cada circuito mostrado en la figura, determine la corriente I a través de la batería.



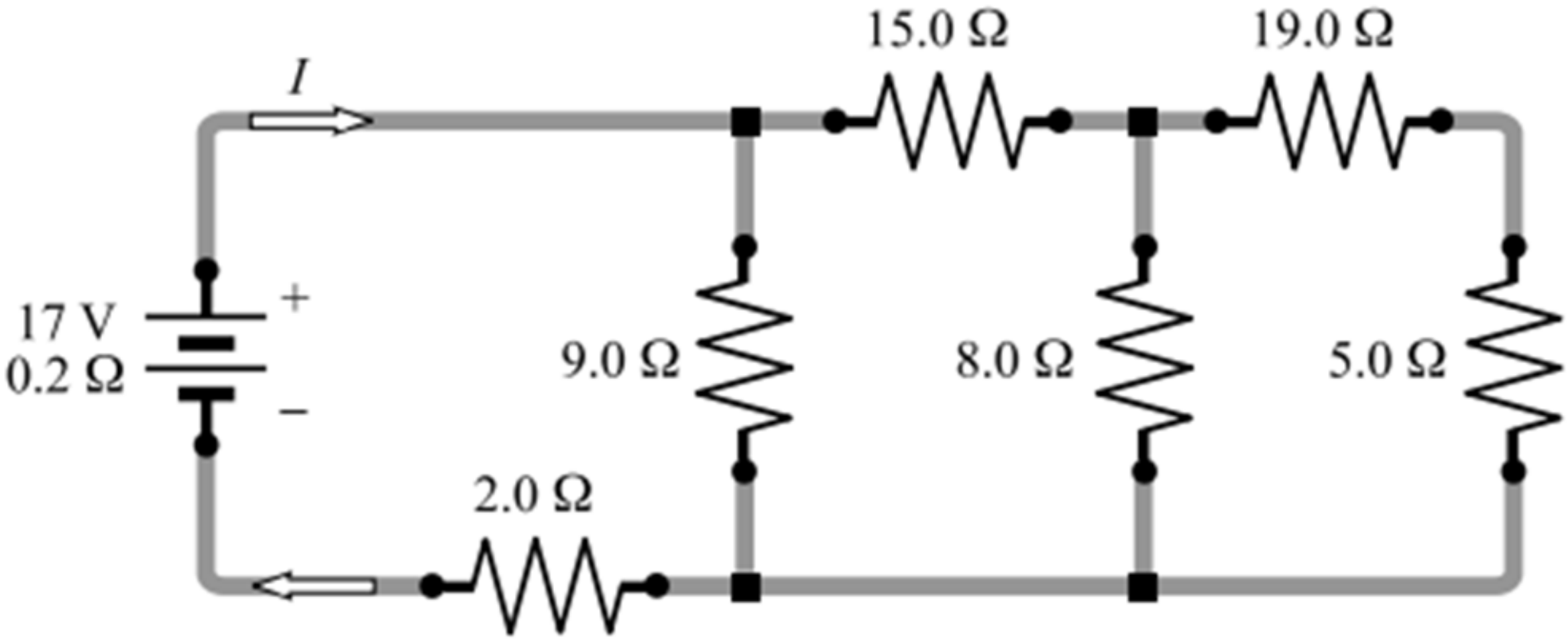
(c)

Proteus 8



(b)

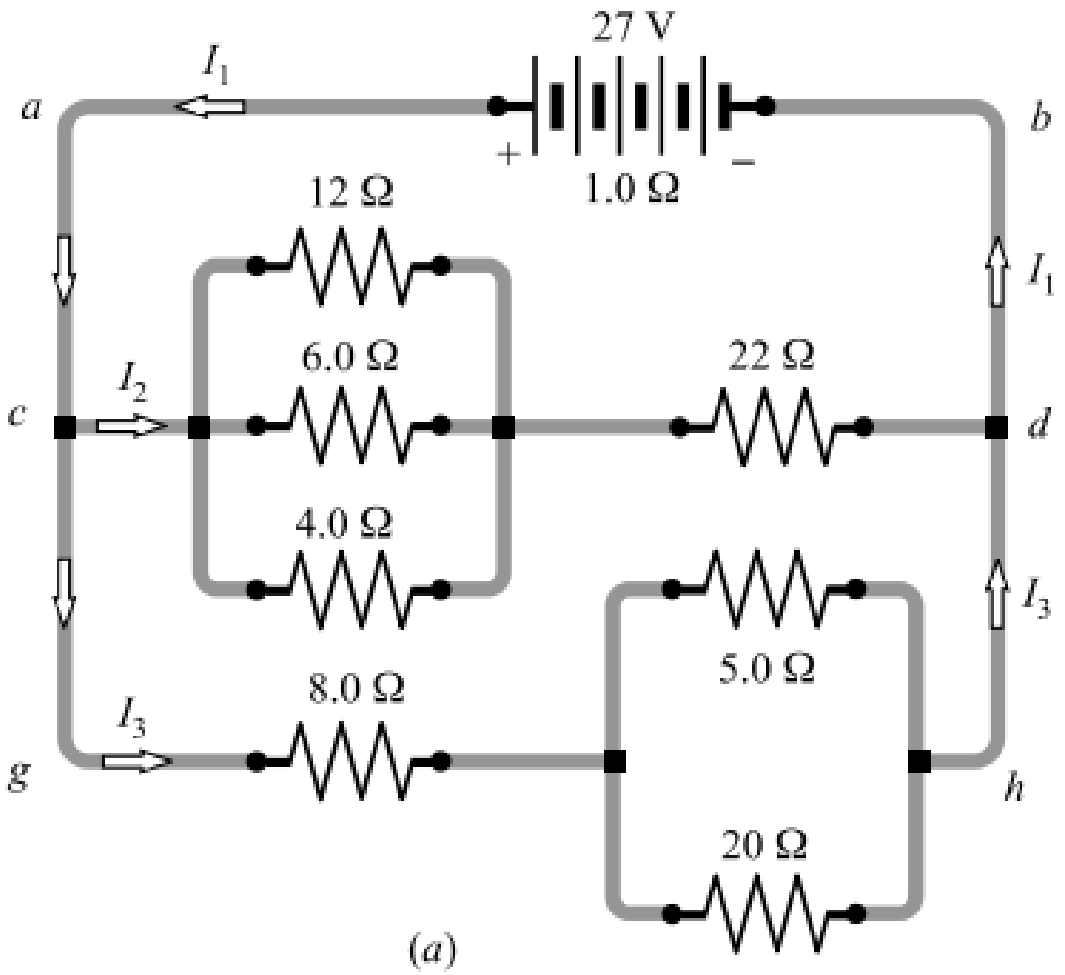
Proteus 9

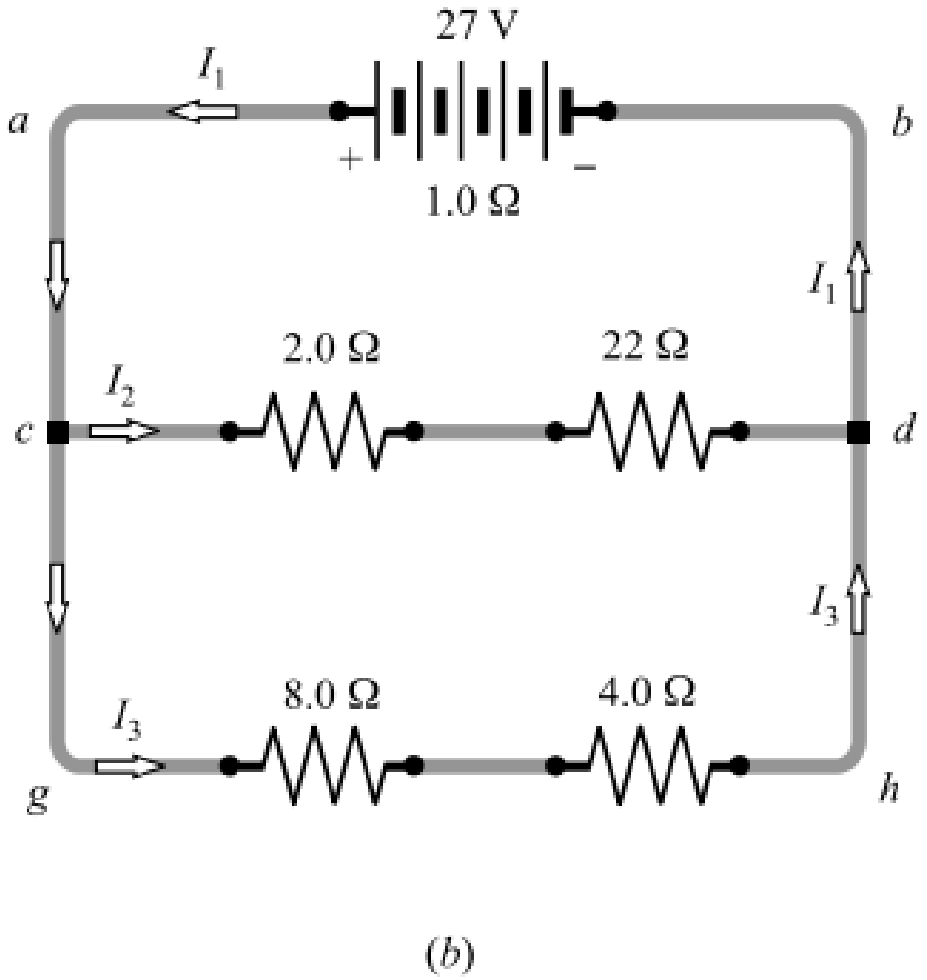


(c)

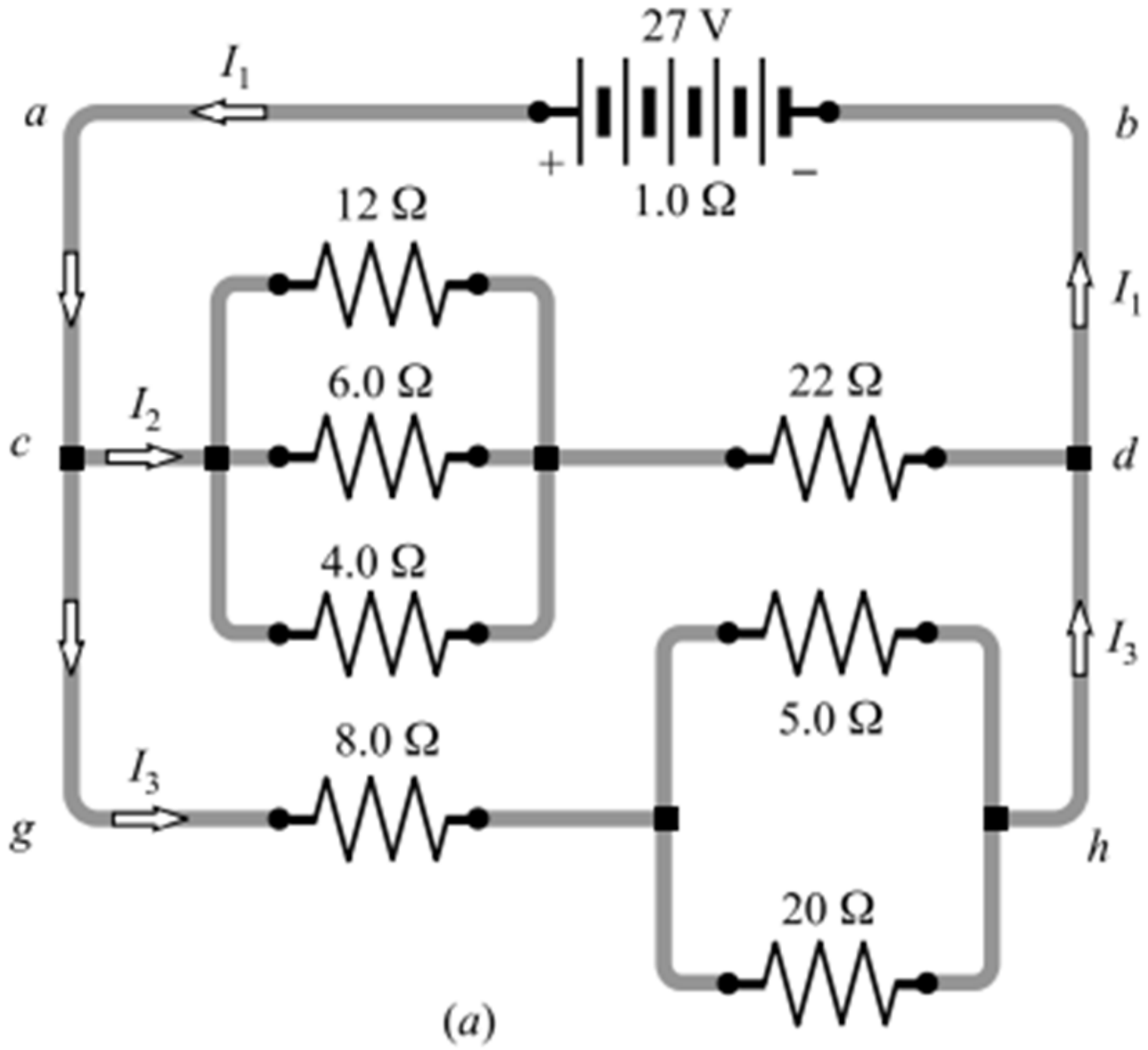
Proteus 10 y 11

- Para el circuito de la figura, encuentre a) I1, I2 e I3; b) la corriente en el resistor de 12 Ω.

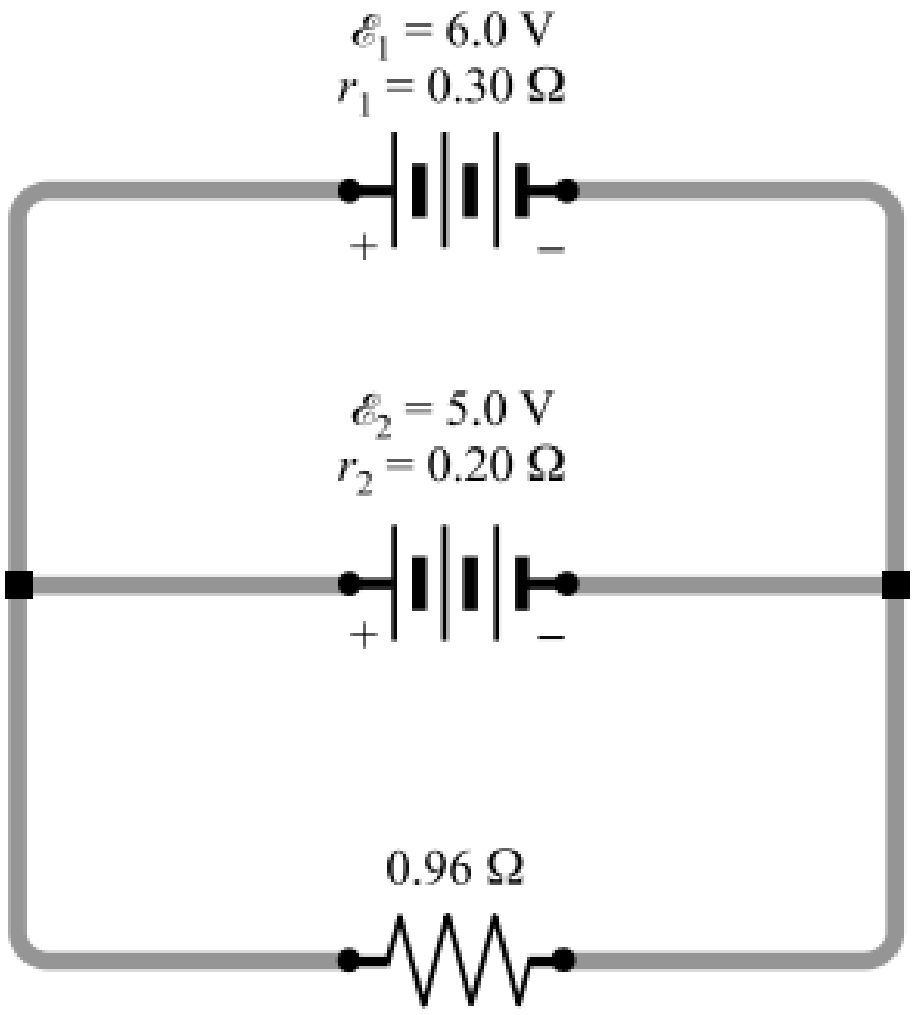




Proteus 12



Para el circuito de la figura, encuentre a) I1, I2 e I3; b) la corriente en el resistor de 12 Ω.

Proteus 13

- Para el circuito que se muestra en la figura, encuentre la corriente en el resistor de 0.96 Ω y los voltajes en las terminales de las baterías.